SHEET 8 – **OPTIMIZATION FOR DATA SCIENCE** SS 2026
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Goal: We examine in this sheet different strategies to set gradient descent step length applied to logistic regression loss function.

Given n I.I.D training data points $\{(x^{(i)}, y^{(i)})\}_{i=1}^{n}$, where each feature vector $x^{(i)} \in \mathbb{R}^d$, and each label $y^{(i)} \in \{0, 1\}$, the regularized loss function $\mathcal{J}_\lambda : \mathbb{R}^d \to \mathbb{R}$ for the task of logistic regression is given by

$$\mathcal{J}_\lambda(\theta) = \sum_{i=1}^{n} \left[\log\left(1 + e^{\theta^\top x_i}\right) - y_i\, \theta^\top x_i\right] + \frac{\lambda}{2}\|\theta\|^2.$$

Exercise 8.1

1. Show that $\mathcal{J}_\lambda$ is μ-strongly convex using two methods, and determine μ.
2. Show that $\mathcal{J}_\lambda$ is L-smooth and determine L.

Exercise 8.2

The following exercise is useful to determine the smoothness constant $L > 0$. Determine $\lambda_{max}(XX^T)$ for $X \in \mathbb{R}^{d\times n}$ as follows:

1. Implement a python function to compute $\lambda_{max}(XX^T)$ via a direct numerical linear algebra routine of your choice.
2. Implement a python function to compute $\lambda_{max}(XX^T)$ via the iterative power method.
3. Compare previous methods' efficiency as d increases: computational time vs accuracy.

Exercise 8.3

Let $f : \mathbb{R}^d \to \mathbb{R}$ be μ-strongly convex and L-smooth with unique minimizer $x^* \in \mathbb{R}^d$. Let $x^0 \in \mathbb{R}^d$, an initial point and consider the gradient descent method with backtracking line search method recalled hereafter

Algorithm 1 *Backtracking line search.*

given the descent direction $-\nabla f(x)$ for f at $x \in \mathbf{dom}\, f$, $\alpha \in (0, 0.5)$, $\beta \in (0, 1)$.
$t := 1$.
while $f(x + t\Delta x) > f(x) + \alpha t \nabla f(x)^T \Delta x, \quad t := \beta t.$

Show that

$$f(x^k) - f(x^*) \leq \left(1 - min\{2\mu\alpha, 2\beta\alpha\mu/L\}\right)^k (f(x^0) - f(x^*)).$$

Exercise 8.4

1. Implement gradient descent method using different fixed stepsizes including $\frac{1}{L}$ and $\frac{2}{\mu+L}$, while estimating L using methods from Exercise 8.2.
2. Implement gradient descent with exact line search as described in Sheet 4, Exercise 4.2.
3. Implement Gradient descent with backtracking step size rule as described in Exercise 8.3.
4. Create a summary comparing abovementioned methods in terms of information required (L, μ..), cost per step, number of iterations, total computational cost. Use a uniform stopping criteria for all methods. Conclude.